Imitation Learning of an LTV-MPC for Real-Time Lane Keeping and Dynamic Obstacle Avoidance

Carmine Maisto, Dario Barbato and Andrea Colapinto

Abstract -

This paper presents a two-stage control architecture addressing lane keeping and obstacle avoidance for autonomous vehicles. In the first stage, a longitudinal Model Predictive Control (MPC) module computes a safe velocity profile by optimizing the vehicle acceleration according to obstacle distance and road friction conditions. The resulting velocity is then used to update a Linear Time-Varying Model Predictive Control (LTV-MPC), which optimizes the lateral dynamics for lane keeping and obstacle avoidance. Subsequently, the optimal solutions generated by the MPC are used as an expert dataset to train a feedforward Neural Network via Imitation Learning, transferring the control policy to a computationally efficient agent. Simulation results demonstrate that both controllers successfully handle obstacle avoidance scenarios under varying friction conditions. Finally, a multidimensional performance comparison is provided, evaluating vehicle trajectories, control inputs, computational load, and cost function trends.

1 Introduction

Over the last decades, the increasing integration of electronics and software in vehicles has enabled the introduction of several active safety systems, such as Anti-lock Braking System (ABS), Electronic Stability Control (ESC), and Adaptive Cruise Control (ACC). Recent advances in perception technologies have paved the way for autonomous driving systems capable of handling complex scenarios. In this context, Model Predictive Control (MPC) [1] has emerged as a standard methodology for addressing combined lane-keeping and obstacle avoidance problems, due to its capability to handle constraints and multivariable systems in a systematic way.

The implementation of nonlinear MPC formulations requires the online solution of non-convex optimization problems, resulting in a significant computational burden that limits the achievable prediction horizon in real-time applications. To reduce this complexity, Linear Time-Varying (LTV) formulations have been proposed in the literature. A notable approach was introduced by Turri et al. [2], where longitudinal and lateral dynamics are decoupled, allowing the problem to be recast as a sequence of convex Quadratic Programs (QP).

In this framework, the longitudinal control layer computes a safe velocity profile by solving an MPC problem that optimizes the vehicle acceleration based on obstacle proximity and friction conditions. The resulting velocity is then used to update the LTV model employed in the lateral MPC, which generates the optimal steering actions for lane keeping and obstacle avoidance. Despite this simplification, the LTV-MPC still requires solving an optimization problem at each sampling instant, which may lead to high computational loads in complex scenarios.

To overcome this limitation, this paper proposes a hybrid control architecture inspired by End-to-End learning paradigms [3]. First, a decoupled LTV-MPC controller is designed as an expert policy generator. Then, the state-action trajectories generated by the MPC are used to train a feedforward Neural Network through Imitation Learning. The objective is to approximate the optimal control policy, enabling real-time inference with a significantly reduced computational cost.

The remainder of the paper is organized as follows: Section 2 introduces the vehicle model; Section 3 describes the proposed methodology, including the LTV-MPC formulation and the Neural Network training; Section 4 presents a detailed comparison between the two approaches; finally, Section 5 provides concluding remarks.

2 Vehicle Dynamics and LTV Formulation

This section presents the vehicle model adopted in this work and the corresponding Linear Time-Varying (LTV) formulation used within the predictive control framework.

2.1 Modelling Assumptions

To obtain a tractable model while preserving the essential vehicle dynamics, the following assumptions are introduced, consistent with typical highway driving conditions:

- Assumption 1 (Low curvature road): The vehicle is assumed to operate on straight or mildly curved roads. Consequently, the orientation error ($\Delta\psi$) and the steering angle remain sufficiently small to justify the small-angle approximations ($\cos(\delta) \approx 1$, $\sin(\delta) \approx \delta$).
- Assumption 2 (Linear tire behavior): The lateral tire forces are approximated using a linear model derived from the Pacejka formulation [4], assuming operation within the linear region. The effective cornering stiffnesses are scaled by the road friction coefficient μ, such that $C_{\alpha f,eff} = C_{\alpha f} \cdot \mu$ and $C_{\alpha r,eff} = C_{\alpha r} \cdot \mu$.

2.2 Continuous-Time Lateral Dynamics

The lateral vehicle behavior is described using an extended dynamic bicycle model [2], shown in Figure 1. The steering angle δ is included in the state vector, allowing the controller to account for its effect on vehicle dynamics.

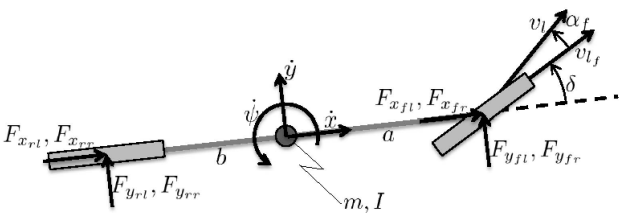


Figure 1. Schematic representation of the extended bicycle model, illustrating the vehicle state variables, geometric parameters (a, b), and the steering input (δ).

The state vector is defined as:

$$x = [v_y,\ r,\ \Delta\psi,\ e_y,\ \delta]^T \tag{1}$$

where v_y is the lateral velocity, r is the yaw rate, $\Delta\psi$ is the heading error, e_y is the lateral displacement with respect to the lane center, and δ is the steering angle.

The system is described by the following linear time-varying continuous-time model:

$$\dot{x}(t) = A_c x(t) + B_c u(t) \tag{2}$$

where $u(t)$ represents the steering control input, i.e., the command generated by the controller that governs the evolution of the steering angle.

The system dynamics are expressed as:

$$\begin{cases} \dot{v}_y = -\frac{C_{\alpha f}^{eff}+C_{\alpha r}^{eff}}{m_{car}v_x} v_y + \frac{C_{\alpha r}^{eff}b-C_{\alpha f}^{eff}a}{m_{car}v_x} r - v_x r + \frac{C_{\alpha f}^{eff}}{m_{car}}\delta \\ \dot{r} = \frac{C_{\alpha r}^{eff}b-C_{\alpha f}^{eff}a}{Iv_x} v_y - \frac{C_{\alpha f}^{eff}a^2+C_{\alpha r}^{eff}b^2}{Iv_x} r + \frac{C_{\alpha f}^{eff}a}{I}\delta \\ \Delta\dot{\psi} = r \\ \dot{e}_y = v_y + v_x \Delta\psi \\ \dot{\delta} = u \end{cases} \tag{3}$$

The continuous-time state matrix A_c is structured as:

$$A_c = \begin{bmatrix} A_{11} & A_{12} & 0 & 0 & A_{15} \\ A_{21} & A_{22} & 0 & 0 & A_{25} \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & v_x & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{4}$$

where:

- $A_{11} = -\frac{C_{\alpha f}^{eff}+C_{\alpha r}^{eff}}{m_{car}v_x}$
- $A_{12} = -v_x + \frac{C_{\alpha r}^{eff}b-C_{\alpha f}^{eff}a}{m_{car}v_x}$
- $A_{15} = \frac{C_{\alpha f}^{eff}}{m_{car}}$
- $A_{21} = \frac{C_{\alpha r}^{eff}b-C_{\alpha f}^{eff}a}{Iv_x}$
- $A_{22} = -\frac{C_{\alpha f}^{eff}a^2+C_{\alpha r}^{eff}b^2}{Iv_x}$
- $A_{25} = \frac{C_{\alpha f}^{eff}a}{I}$

The input matrix is defined as:

$$B_c = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \end{bmatrix}^T \tag{5}$$

The physical parameters used are $m_{car} = 2050$ kg, $I = 3344$ kg·m^2, $a = 1.2$ m, and $b = 1.6$ m.

2.3 Continuous-Time Longitudinal Dynamics

The longitudinal dynamics are modeled separately to regulate vehicle speed and enable safe braking in the presence of obstacles.

The longitudinal state is defined by the vehicle position X and longitudinal velocity v_x. The dynamics are:

$$\begin{cases} \dot{X}(t) = v_x(t) \\ \dot{v}_x(t) = a_x(t) \end{cases} \tag{6}$$

where $a_x(t)$ is the longitudinal acceleration, representing the combined action of throttle and braking.

The longitudinal velocity v_x is used to update the lateral LTV model at each time step, making the system explicitly time-varying.

2.4 LTV Formulation and Discretization

Since the matrix A_c explicitly depends on the longitudinal velocity v_x, the system is inherently time-varying. To formulate the MPC problem as a sequence of convex Quadratic Programs (QPs), the model is updated and discretized at each sampling instant ($T_s = 0.1$ s).

At each time step, the current velocity v_x is measured and constrained to remain above a minimum threshold of 0.1 m/s in order to avoid numerical singularities. The continuous-time model is then discretized using the matrix exponential, yielding the discrete-time matrices A_{dyn} and B_{dyn}.

This approach allows the prediction model used within the optimizer to adapt online to the current vehicle operating conditions, including acceleration and braking phases.

2.5 Safety Constraints and Operating Environment

Safety requirements and physical limitations of the vehicle are incorporated into the predictive control problem through linear constraints on both states and inputs.

- Steering input limits: The steering command is bounded to account for actuator limitations, such that $u \in [-0.6,\ 0.6]$.

- Longitudinal acceleration limits: The admissible longitudinal acceleration depends on the available tire-road friction coefficient μ. The bounds are defined as $a_{min} = -0.8\,\mu g$ (maximum braking) and $a_{max} = 0.4\,\mu g$ (maximum traction).
- Lane boundaries: The lateral position of the vehicle is constrained within the road limits. Considering a fixed-width lane, the admissible region is defined as $y \in [-8, 8]$ meters.

3 Methodology

This section describes the framework used to compare the performance of a Neural Network (NN) controller against a Linear Time-Varying Model Predictive Controller (LTV-MPC) for autonomous navigation.

3.1 Decoupling Strategy

Following the approach proposed in [2], the control architecture adopts a decoupled structure between longitudinal and lateral dynamics. The control problem is divided into two interacting modules: a longitudinal module responsible for speed regulation and a lateral LTV-MPC responsible for trajectory tracking and obstacle avoidance.

The longitudinal dynamics are modeled as a discrete-time integrator with sampling time $T_s = 0.1$ s:

$$v_{x,k+1} = \max(v_{x,k} + a_{x,k}T_s,\ 0.1) \tag{7}$$

At each time step, the longitudinal module determines a safe reference velocity based on obstacle proximity, lateral clearance, and friction conditions. In particular, when an obstacle is detected within a predefined distance, a safe velocity is computed as:

$$v_{safe} = dist_{obs}\sqrt{\frac{\mu g k}{\Delta y_{min}}} \tag{8}$$

where g is the gravitational acceleration, $k = 0.3$ is a safety factor, and Δy_{min} represents the minimum lateral displacement required to clear the obstacle:

$$\Delta y_{min} = \max\left(\min\left(|y - y_{sup}|,\ |y - y_{inf}|\right),\ 0.5\right) \tag{9}$$

The resulting velocity is saturated between a minimum threshold of 5.0 m/s and the desired cruise speed $v_{x,target}$ to obtain the target velocity.

A longitudinal MPC is then used to compute the acceleration command a_x required to track this target velocity while satisfying friction-dependent acceleration limits.

The resulting longitudinal velocity v_x is used to update the lateral model. The lateral dynamics are described by a Linear Time-Varying system whose matrices depend on v_x. At each time step, the continuous-time model is updated and discretized to obtain $A_{dyn,k}$ and $B_{dyn,k}$.

3.2 LTV-MPC Formulation

The lateral controller is implemented as a Model Predictive Controller with a prediction horizon of $N_{horizon} = 30$.

At each sampling instant, the controller solves a Quadratic Program (QP) to determine the optimal sequence of steering commands:

$$\begin{aligned} \min_{U} \quad & \sum_{k=1}^{N_{horizon}} \left(X_k^T Q X_k + U_k^T R U_k\right) \\ \text{s.t.} \quad & X_{k+1} = A_{dyn,k}X_k + B_{dyn,k}U_k \\ & u_{min} \le U_k \le u_{max} \\ & y_{min,dyn}(k) \le Y_k \le y_{max,dyn}(k) \end{aligned} \tag{10}$$

where $Q = C^T C$ and $R = R_\rho$ are weighting matrices.

Obstacle avoidance is enforced through dynamic lateral constraints. At each time step, the environment is processed to extract the distance to the next obstacle and the corresponding admissible lateral bounds, defining a feasible driving corridor.

To handle the non-convexity of obstacle avoidance (e.g., overtaking from different sides), multiple candidate constraint sets are evaluated and the solution with the minimum cost is selected.

Finally, the system dynamics are propagated over the prediction horizon to obtain a dense QP formulation, enabling efficient numerical solution at each sampling instant.

3.3 Dataset Generation via Imitation Learning

To train the neural surrogate model, an End-to-End Imitation Learning (Behavioral Cloning) approach was employed. A total of $N = 2500$ independent simulations were executed with randomized friction coefficients and target velocities.

At each time step i, the environment provided a feature vector $\phi_i \in \mathbb{R}^{11}$:

$$\begin{aligned} \phi_i = \big[X_i^T,\ v_{curr},\ dist_{next_obs},\ y_{upper}, \\ y_{lower},\ \mu,\ v_{x,target}\big]^T \end{aligned} \tag{11}$$

These features include the vehicle state, current velocity, obstacle information, road friction coefficient, and desired speed.

The corresponding target outputs were obtained from the MPC controller. In particular, the steering control input and the realized longitudinal acceleration were recorded:

$$Y_{target,i} = [u_i,\ a_{x,i}]^T \tag{12}$$

3.4 Neural Network Architecture and Training

The surrogate controller was implemented as a fully connected Multi-Layer Perceptron (MLP). The network consists of an input layer (11 features), two hidden layers with 40 and 20 neurons respectively, and an output layer with 2 neurons corresponding to the control actions.

The network was trained offline using backpropagation to minimize the Mean Squared Error (MSE) between predicted and

target control actions over the dataset. The trained network provides a fast approximation of the control policy learned from the MPC.

3.5 Evaluation Metrics and Cost Function

To compare the performance of the NN surrogate and the MPC controller, an evaluation framework was defined based on cumulative cost indices.

At each time step, the lateral cost is defined as:

$$J_{lat,k} = Q_{lat} Y_k^2 + R_{lat} U_k^2 + J_{pen,k} \tag{13}$$

where the penalty term accounts for violations of safety margins. A fourth-order penalty function is used:

$$J_{pen,k} = W_{margin} \cdot (\Delta y_{inv,k})^4 \tag{14}$$

The longitudinal cost is defined as:

$$J_{lon,k} = Q_{lon}(v_{x,k} - v_{x,target,k})^2 + R_{lon} a_{x,k}^2 \tag{15}$$

The cumulative lateral and longitudinal costs are averaged over the simulation length, and the total average cost is computed as:

$$J_{tot} = J_{lat} + J_{lon} \tag{16}$$

To account for numerical discretization effects, a small tolerance $\epsilon_{num} = 0.05$ m is introduced when evaluating constraint violations.

4 Results and Discussion

To rigorously validate the proposed Neural Network (NN) surrogate controller, a comprehensive testing framework was established in the simulation environment. This section analyzes the performance across two distinct scenarios: a high-friction surface ($\mu = 0.8$, simulating dry asphalt) and a challenging low-friction surface ($\mu = 0.5$, simulating wet or snowy asphalt). These benchmarks evaluate the surrogate's ability to generalize complex vehicular physics learned via Imitation Learning.

The road environment consists of a 300 m straight roadway populated with 5 spatially synchronized static obstacles. The global target velocity was set to $v_{x_target} = 15$ m/s (54 km/h) across all tests. Objective performance is quantified by the independent Referee framework.

4.1 Scenario A: High-Friction Surface ($\mu = 0.8$)

This benchmark evaluates performance in near-ideal driving conditions. The primary objective is to verify that the NN can accurately replicate the expert's optimal spatial and temporal behavior.

4.1.1 Trajectory and Speed Tracking ($\mu = 0.8$)

The 2D spatial trajectory analysis (Figure 2, top) reveals near-perfect alignment between the LTV-MPC expert and the neural surrogate. The longitudinal velocity profile (Figure 2, bottom) confirms that both controllers decrease the speed from 54 km/h to a safe threshold of approximately 42 km/h before the obstacles, prioritizing energy efficiency when high grip is available.

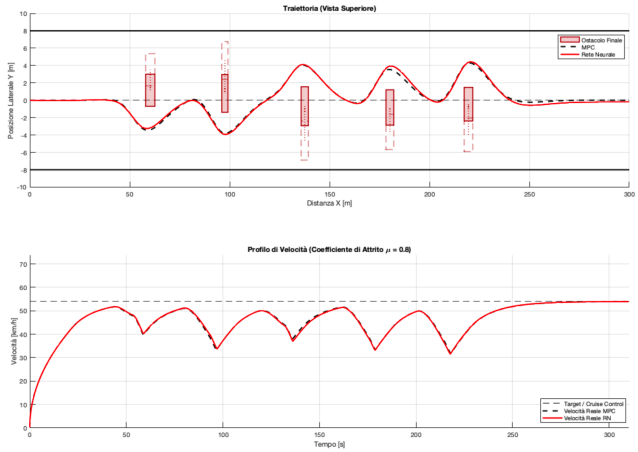


Figure 2. Scenario A ($\mu = 0.8$): Comparison of the 2D spatial trajectory and longitudinal velocity profiles.

4.1.2 Actuator and Safety Analysis ($\mu = 0.8$)

The control effort analysis (Figure 3) demonstrates that the NN maps the MPC commands efficiently, fully respecting the steering and acceleration physical boundaries without introducing high-frequency chattering.

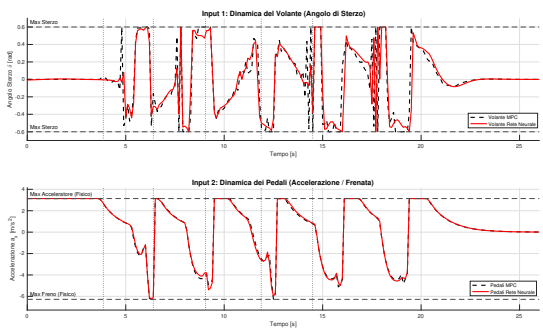


Figure 3. Scenario A ($\mu = 0.8$): Actuator comparison of the steering angle and longitudinal acceleration.

Crucially, the clearance metric (Figure 4) confirms that the neural agent closely tracks the mathematical safety margin optimized by the MPC (1.9 m),ensuring that the physical collision threshold (1.0 m) is not violated in the tested scenarios.

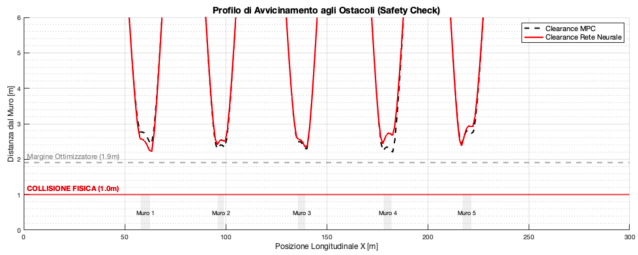
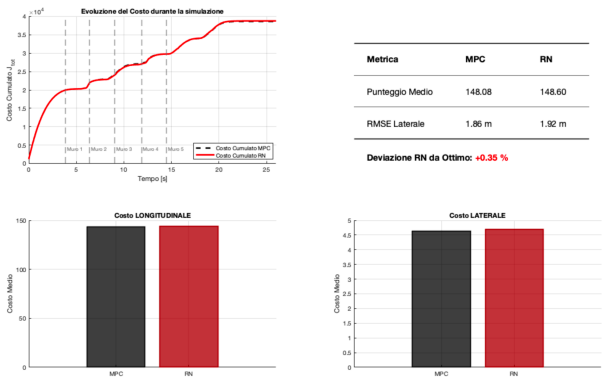


Figure 4. Scenario A ($\mu = 0.8$): Clearance analysis confirming safe obstacle avoidance.

4.1.3 Cost and Computational Performance ($\mu = 0.8$)

The high-friction scenario yields very accurate cost reproduction with a deviation of the NN from the optimum of only +0.35% (Figure 5). The NN surrogate maintains a Root Mean Square Error (RMSE) for lateral deviation that matches the MPC optimum.



Metrica	MPC	RN
Punteggio Medio	148.08	148.60
RMSE Laterale	1.86 m	1.92 m

Deviazione RN da Ottimo: +0.35 %

Figure 5. Scenario A ($\mu = 0.8$): KPI Dashboard illustrating the optimal cumulative cost mimicry.

Regarding execution time (Figure 6), the NN demonstrates its main advantage: a constant, bounded computational footprint ($\approx$ 5 ms) that entirely avoids the variance and peaks exhibited by the MPC QP solver.

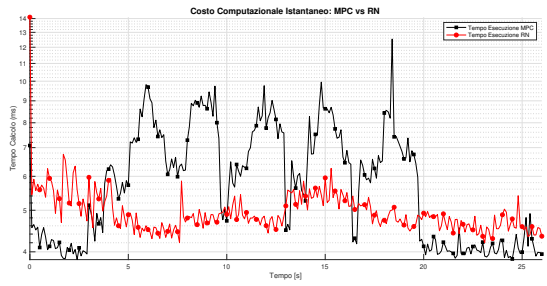


Figure 6. Scenario A ($\mu = 0.8$): Instantaneous computational time per step (logarithmic scale).

4.2 Scenario B: Low-Friction Surface ($\mu = 0.5$)

By lowering the friction coefficient to $\mu = 0.5$, the vehicle's dynamic boundaries contract significantly, providing a severe stress-test for the surrogate's robustness.

4.2.1 Trajectory and Speed Tracking ($\mu = 0.5$)

With reduced grip, both controllers adopt wider lateral avoidance paths (Figure 7). The NN correctly generalizes the physical relationship between grip and safe avoidance speed (Eq. **??**), applying strong braking actions to reach a lower safety threshold of approximately 25 km/h before the evasive maneuvers.

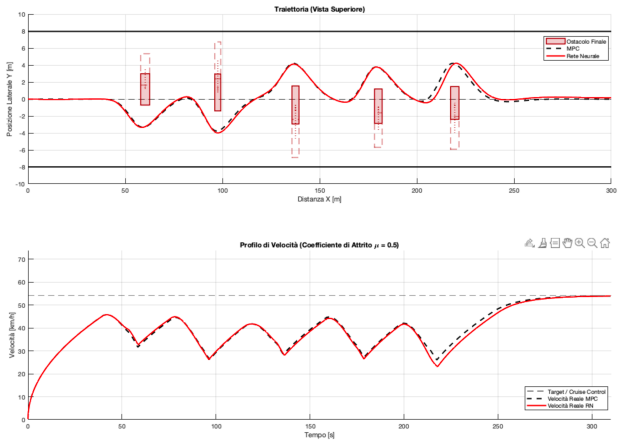


Figure 7. Scenario B ($\mu = 0.5$): Stress-test on wet asphalt. The models proactively adapt by aggressively reducing velocity to 25 km/h.

4.2.2 Actuator and Clearance Analysis ($\mu = 0.5$)

Figure 8 demonstrates that the NN correctly saturates the braking commands at the physically available friction limit (-4.0 m/s^2) just prior to the evasive maneuvers, maximizing the limited grip.

The clearance plot (Figure 9) confirms that safety is maintained in the tested scenarios. Like the previous case, with low-friction surface too, the NN is capable of tracking the MPC trajectory ensuring that the physical condition is avoided, since the vehicle keeps itself at least 1.9 m (MPC margin) from the obstacle.

4.2.3 Cost and Computational Performance ($\mu = 0.5$)

Even under extreme stress, the KPI dashboard (Figure 10) confirms highly effective mimicry. The normalized cumulative cost score represents a minor deviation (e.g., +2.45%) from the expert's mathematical optimum, with an almost identical lateral RMSE (1.88 m vs 1.85 m).

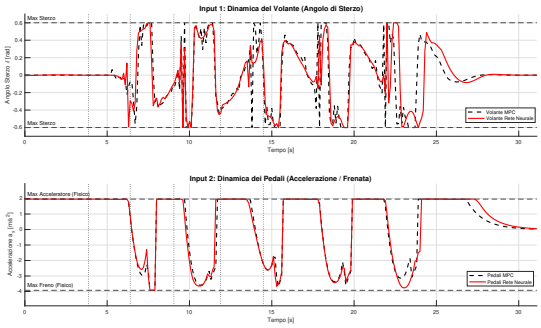


Figure 8. Scenario B ($\mu = 0.5$): Actuator stress-test showing deep braking saturation.

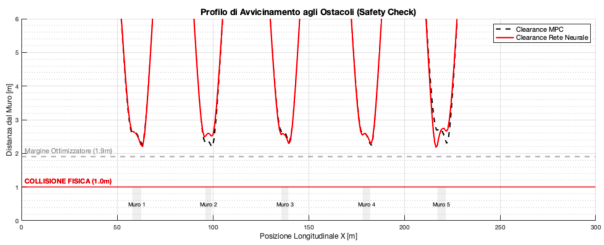
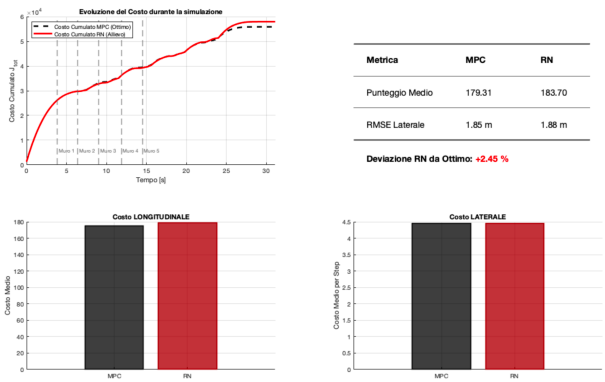


Figure 9. Scenario B ($\mu = 0.5$): Clearance stress-test. The absolute physical threshold is fully respected.



Metrica	MPC	RN
Punteggio Medio	179.31	183.70
RMSE Laterale	1.85 m	1.88 m

Deviazione RN da Ottimo: +2.45 %

Figure 10. Scenario B ($\mu = 0.5$): KPI Dashboard synthesized from the low-friction stress-test.

Finally, the computational analysis (Figure 11) solidifies the neural surrogate's superiority for real-time applications. While the traditional MPC execution time spikes dangerously (nearly 10 ms) during the complex low-friction maneuvers, the NN infers the control actions deterministically via feedforward matrix multiplications, remaining flat and bounded.

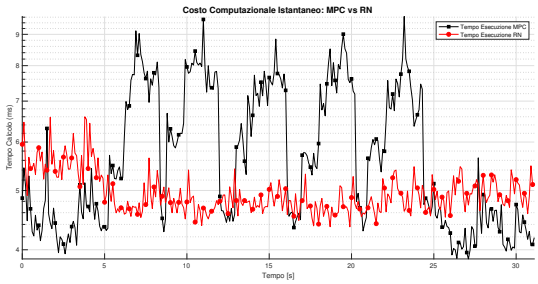


Figure 11. Scenario B ($\mu = 0.5$): Logarithmic scale visualizes the severe and stable computational reduction achieved by the NN.

5 Conclusions

This paper presented a comparative study between a decoupled Linear Time-Varying Model Predictive Control (LTV-MPC) and a deep Neural Network (NN) surrogate for autonomous lane keeping and obstacle avoidance. Through End-to-End Imitation Learning, the optimization-based predictive policy of the MPC was approximated by a highly efficient feedforward neural architecture.

The simulation framework, evaluated under both high ($\mu = 0.8$) and low ($\mu = 0.5$) friction conditions, demonstrated strong performance of the neural surrogate across different driving conditions. The NN successfully replicated the expert's spatial trajectories and adaptive braking behavior, yielding a minimal performance cost deviation and operating without violating the physical collision thresholds in the tested scenarios.

The most significant advantage of the neural surrogate lies in its computational efficiency. While the traditional LTV-MPC solver exhibits execution times with noticeable variability and peaks during complex maneuvers, the NN infers control actions through deterministic feedforward computations. This results in a bounded and predictable execution time, which is beneficial for real-time applications.

Consequently, this study suggests the practical viability of deploying advanced predictive control strategies on resource-constrained automotive Electronic Control Units (ECUs).

Future research will focus on Hardware-in-the-Loop (HIL) validation of the neural controller on automotive microcontrollers. Additionally, the training dataset will be extended to include dynamic environments with moving obstacles and variable-curvature road geometries.

References

[1] Paolo Falcone, Francesco Borrelli, Jahan Asgari, Hongtei Eric Tseng, and Davor Hrovat. Predictive active steering control for autonomous vehicle systems. *IEEE Transactions on control systems technology*, 15(3):566–580, 2007.

[2] Valerio Turri, Ashwin Carvalho, Hongtei Eric Tseng, Karl Henrik Johansson, and Francesco Borrelli. Linear model predictive control for lane keeping and obstacle avoidance on low curvature roads. In *16th International IEEE Conference on Intelligent Transportation Systems (ITSC 2013)*, pages 378–383, 2013. doi:10.1109/ITSC.2013.6728261.

[3] Mariusz Bojarski, Davide Del Testa, Daniel Dworakowski, Bernhard Firner, Beat Flepp, Prasoon Goyal, Lawrence D Jackel, Mathew Monfort, Urs Muller, Jiakai Zhang, et al. End to end learning for self-driving cars. *arXiv preprint arXiv:1604.07316*, 2016.

[4] Hans Pacejka. *Tire and vehicle dynamics*. Elsevier, 2012.